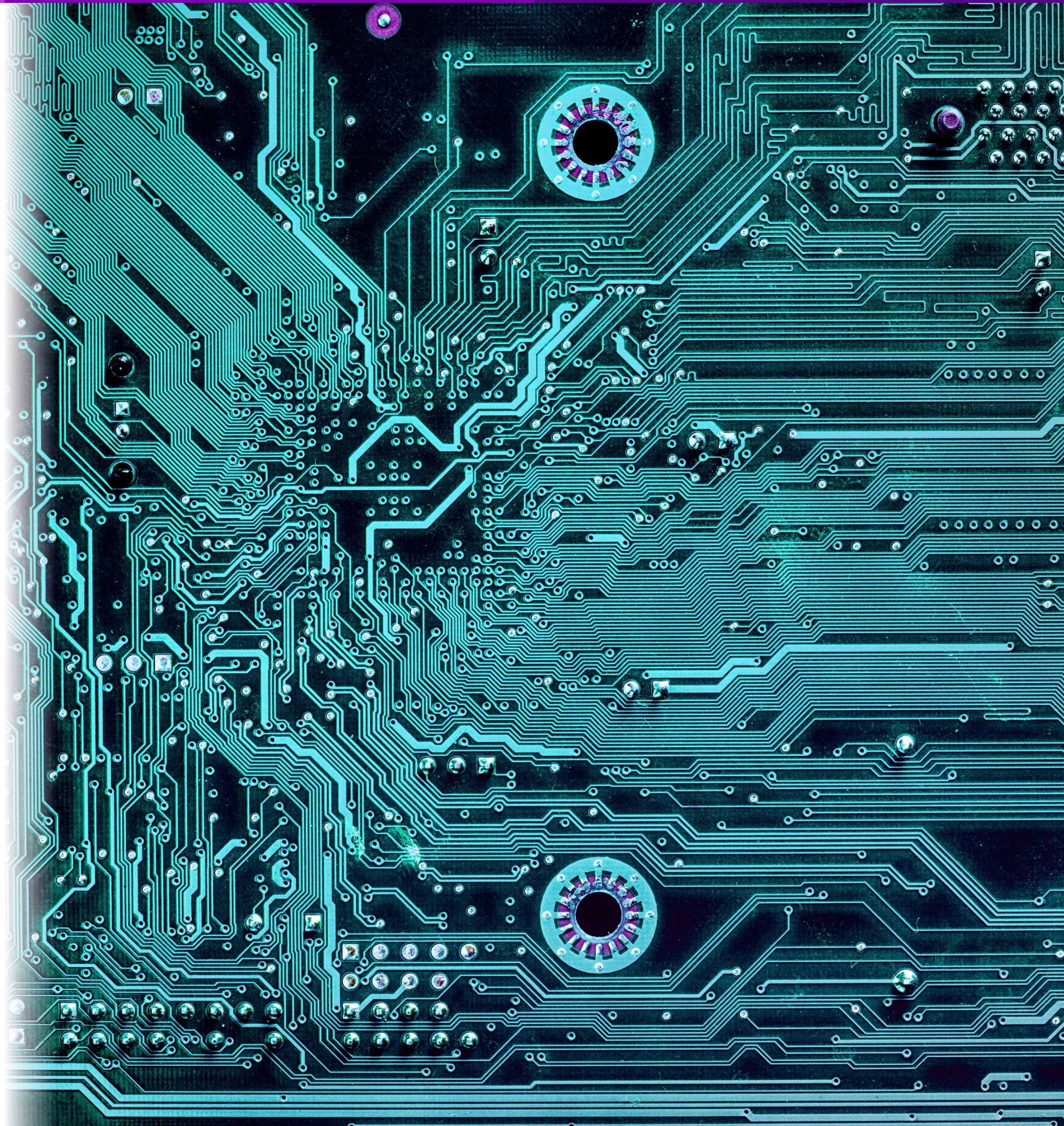
Module 1

Signals and Interfaces

MTE 220 – Sensors and Instrumentation



UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Topics

1 – Signals and Interfaces

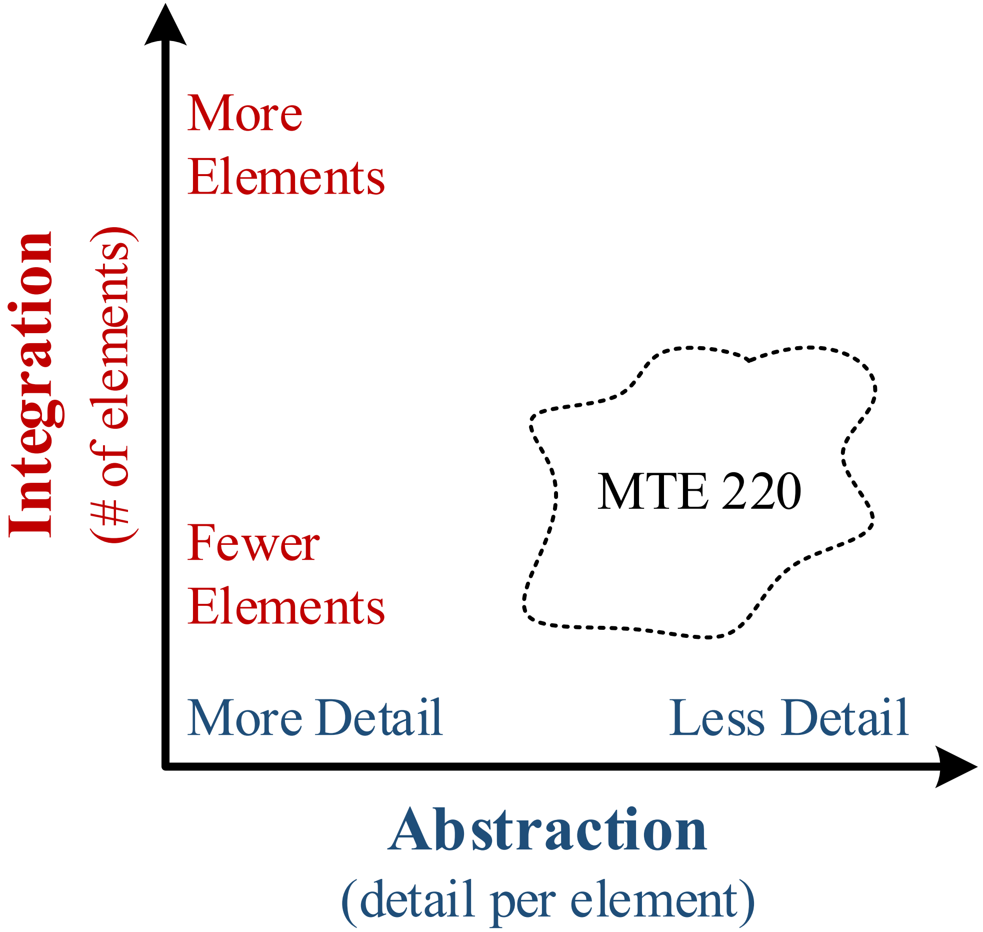
2 – New Devices and Circuits

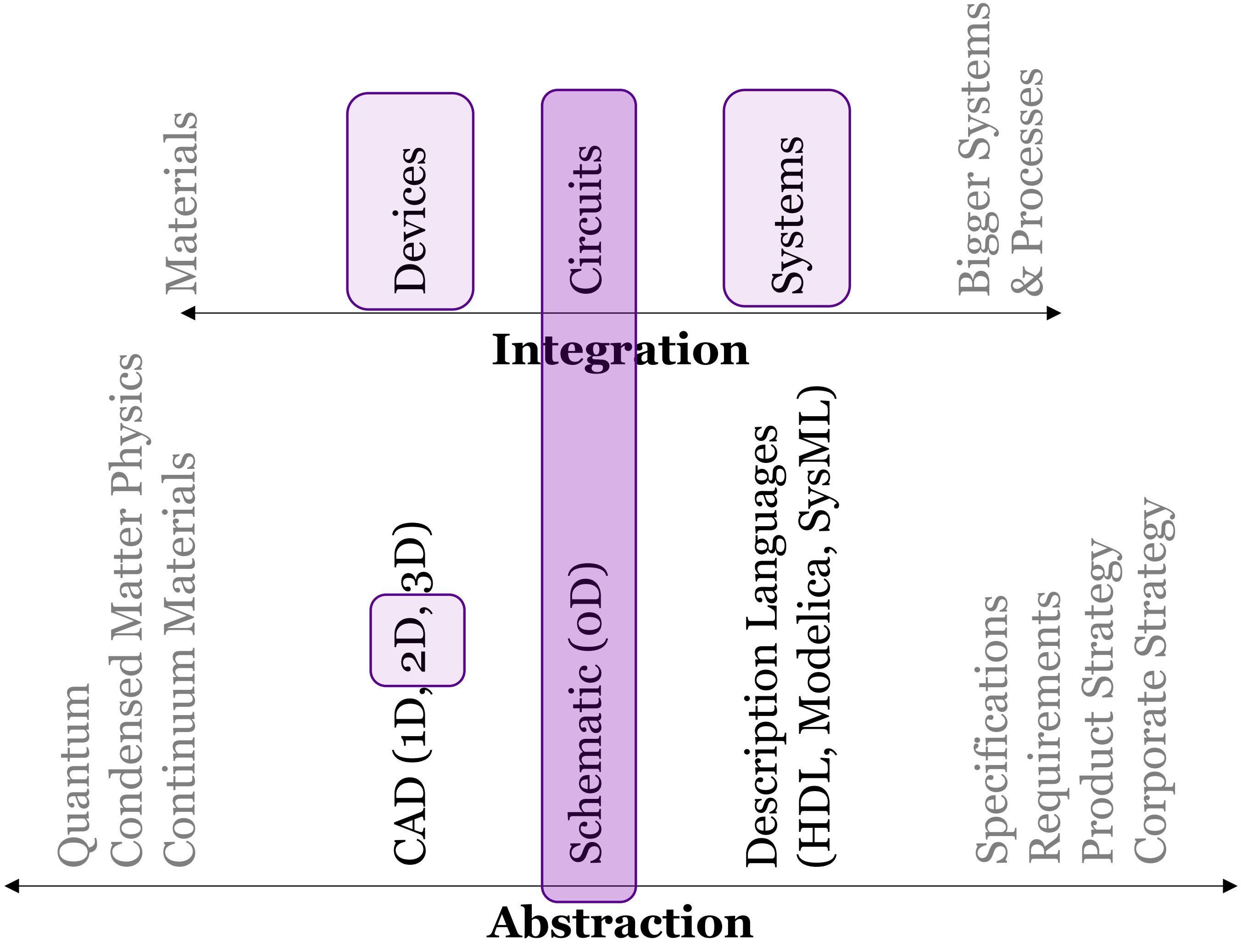
3 – Sensors

4 – Interfacing and Design

- 1.1 – Signal Basics
- 1.2 – Interfaces and Loading
- 1.3 – DC Operating Point and Signal Variation

Preamble: Integration and Abstraction





1.1 – Signal Basics

A Signal Is...

- A measurable variation that carries information
  - Variation may occur in time, space, frequency, or another state variable
  - Variation may be represented electrically, mechanically, optically, digitally, or otherwise
- In MTE 220: A voltage or current that represents information

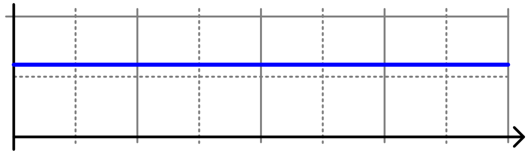
Force $F(t)$ → Strain Gauge → Wheatstone Bridge → $V(t)$ → Amplifier → ADC → MCU → Motor Controller

Temperature $T(t)$ → Thermistor → Voltage Divider → $V(t)$ → Filter → ADC → MCU → Fan Controller

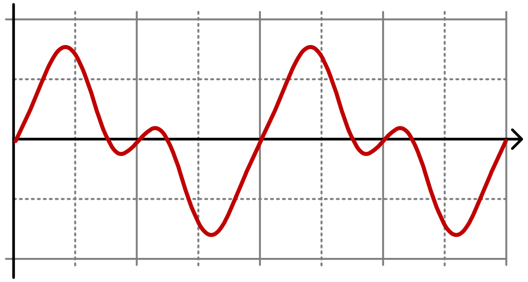
Light $E(t)$ → Photodiode → TIA → $V(t)$ → Schmitt Trigger → ADC → MCU → Door Actuator

AC and DC

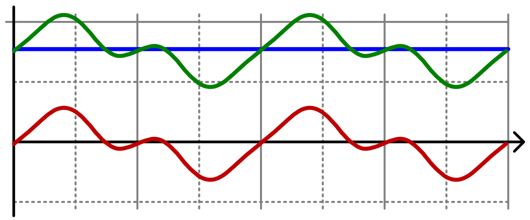
- Direct Current (DC)
  - Original Meaning: Current and applied voltage do not change directions, but may fluctuate



  - Modern Meaning: Current or voltage is constant (also called “zero frequency component”)
- Alternating Current (AC)
  - Original Meaning: Current and applied voltage changes directions



  - Modern Meaning: Current or voltage is time-varying and has no constant offset (also called “non-zero frequency content”)
- We apply this terminology to any signal, regardless of physical meaning



  - E.g., “The total signal is the sum of the DC offset and AC components”

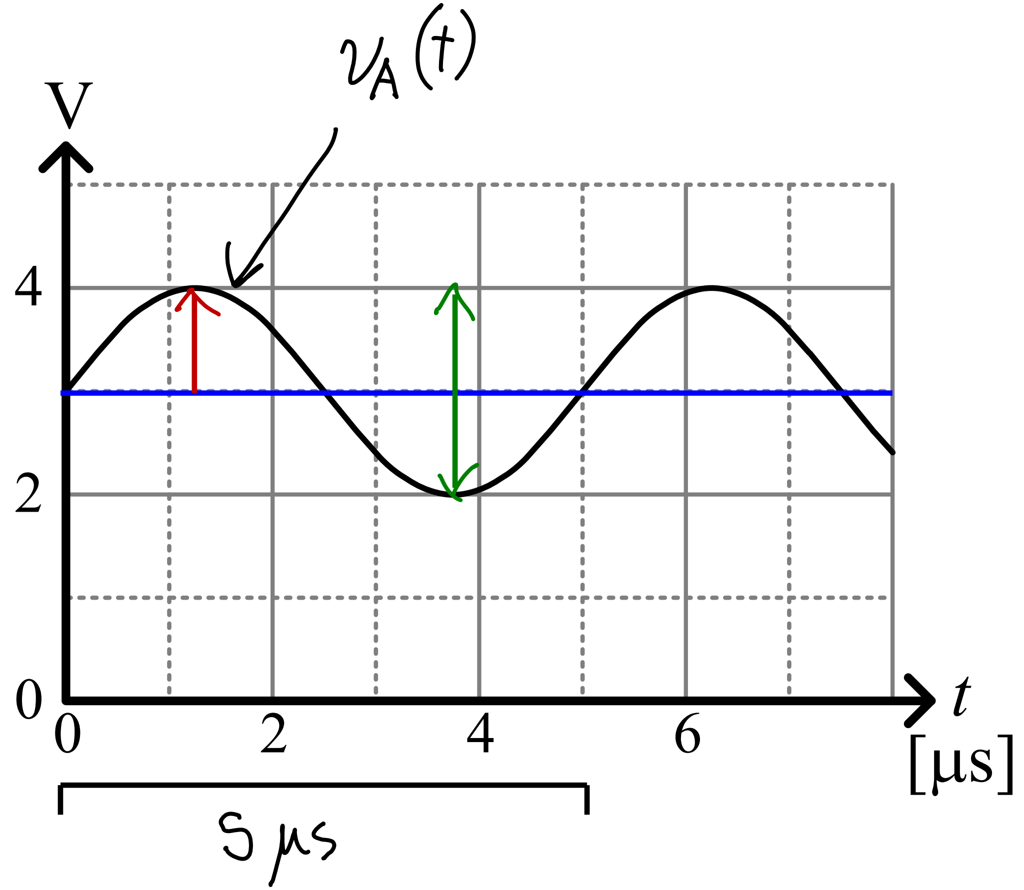
Waveform Nomenclature

- From MTE 120:

Parameter	Value
V_{DC}	3 V
V_{pk}	1 V
$V_{pk\text{-}pk}$	2 V
T_o	5 µs
f_o	200 kHz
ω_o	1.257 Mrad/s

$= 1/T_o$

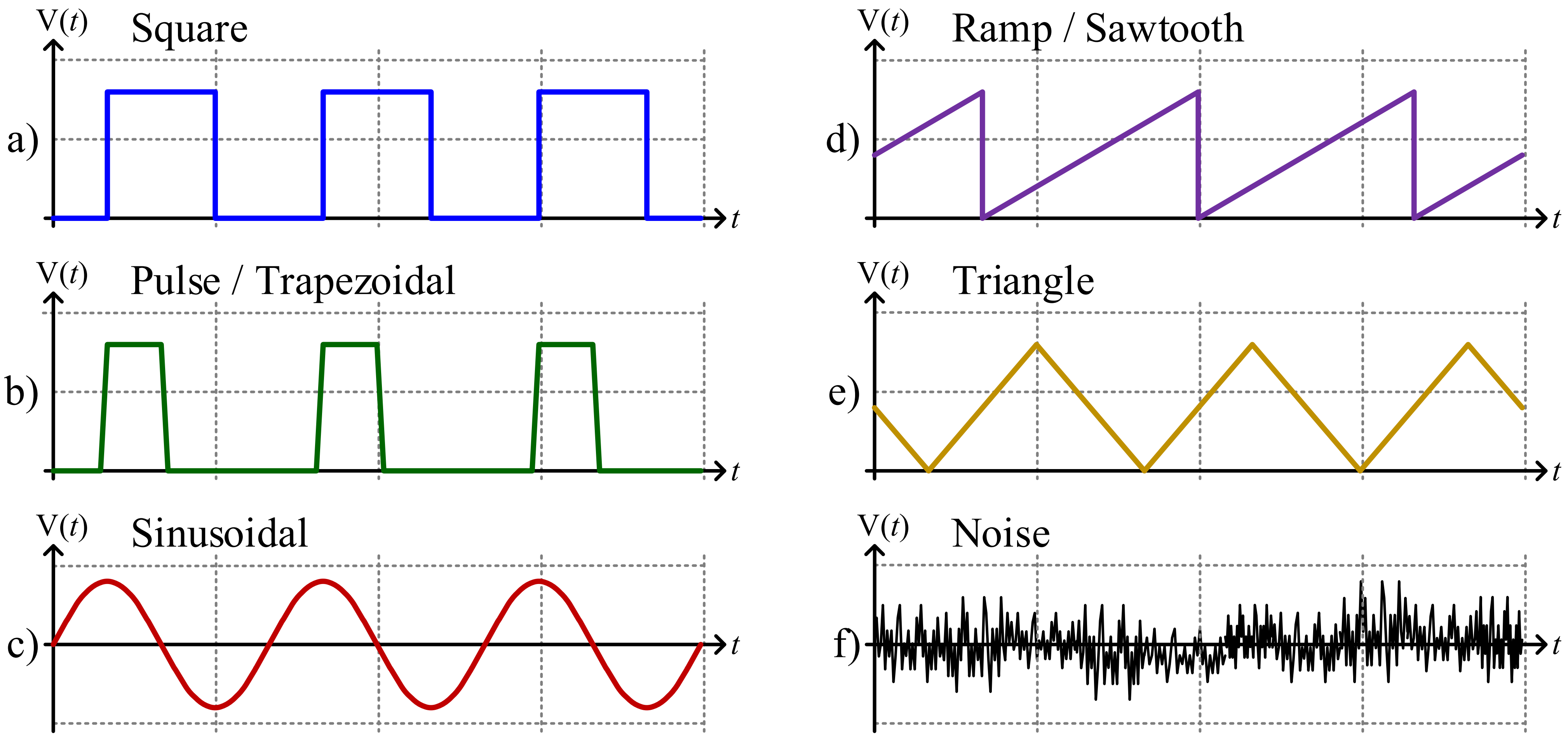
$= 2\pi f_o$



- Analytically, $v(t) = 3 + \sin(2\pi \times 200\text{k} \times t)$
- Nomenclature: $v_A(t) = V_A + v_a(t)$

total signal = DC offset + AC component

Common Waveforms



Linearity and Signals

We care so much about sinusoidal response because:

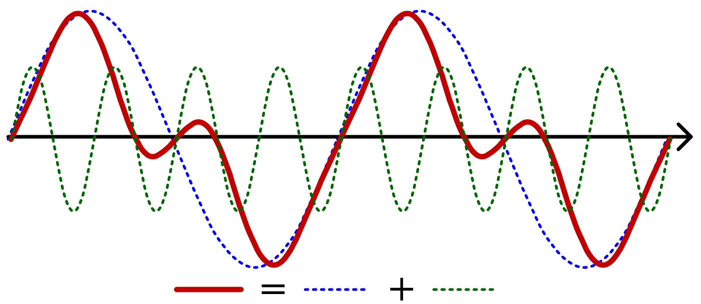
1. **Any signal is a sum of sine waves**

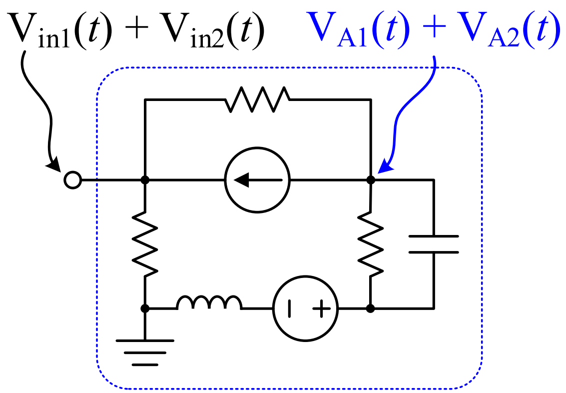
2. **Superposition applies to linear systems**

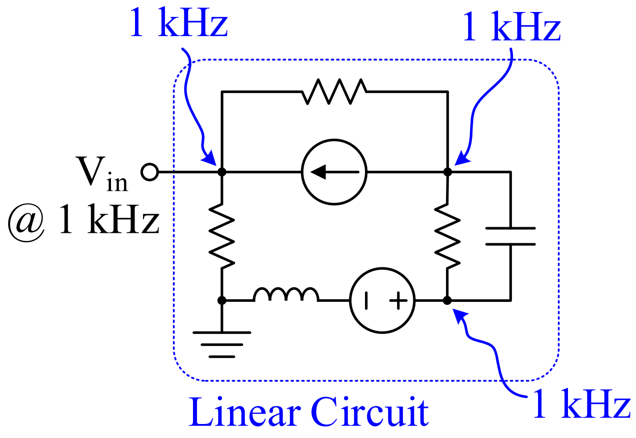
3. **Linear operations don't change the frequency**

So, we tackle one frequency at a time in sinusoidal analysis

We can add the responses together to get the total response

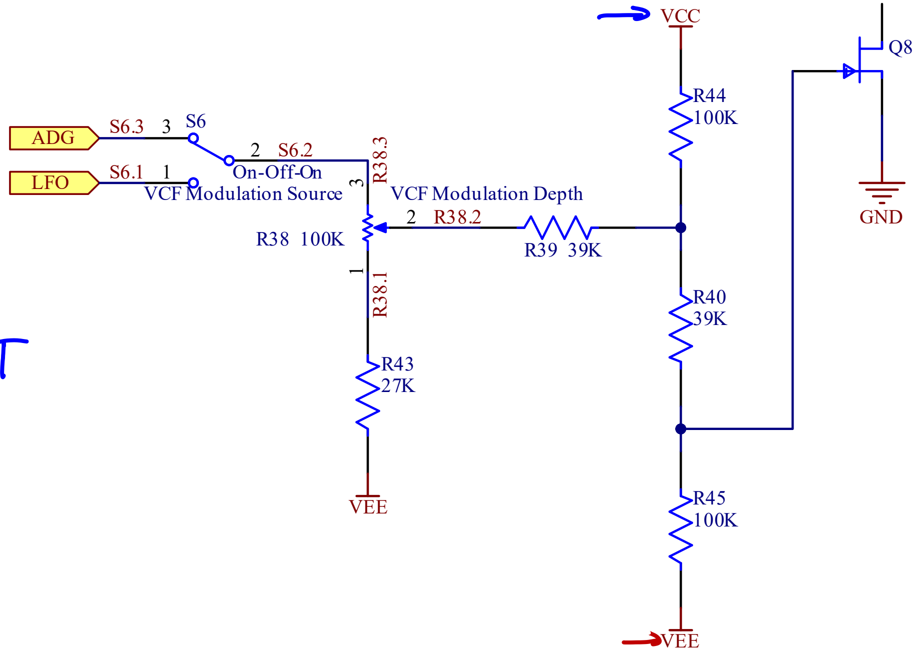


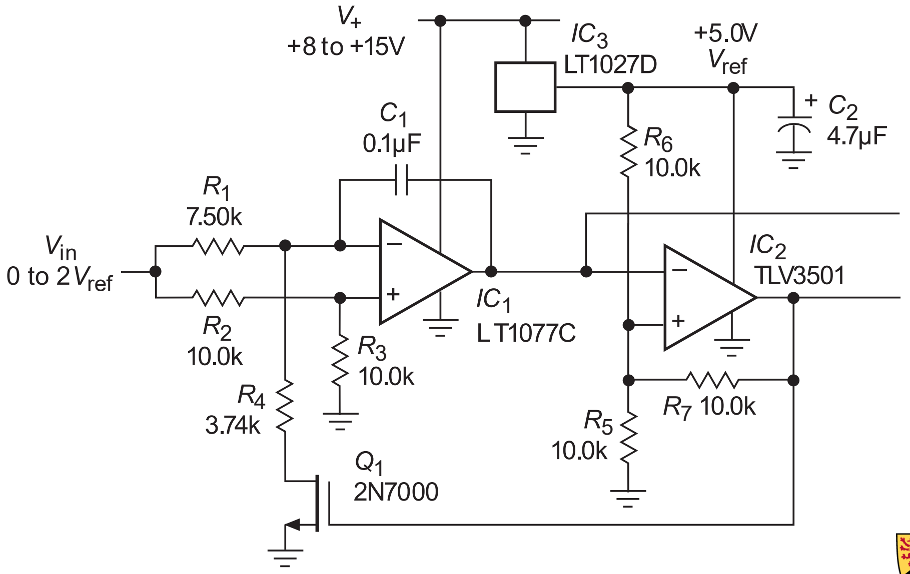




Voltage Rails

- Top Rail = max working voltage
  - V_{DD} or V_{CC}, or explicit: “+5 V”, “+3.3 V” + BATT
- Bottom Rail = min working voltage
  - V_{SS}, V_{EE}, or explicit: “−5 V”, “−9 V”
- Ground = 0 V; may also be the bottom rail
- Signals **must** remain between bottom and top rails (typically)
- Exceeding the rails can damage components





LTSpice Modeling Challenge

- Create voltage sources that output the waveforms on Slide 21
  - 1u = 1 micro
  - 1Meg = 1 million
  - 1m = 1M = 1 milli
  - This button  lets you configure a “transient” (time-domain) simulation
- What are the challenges with square and ramp?
- What are the challenges with triangle?
- How did you make a noise source?
- Can you recreate the plot layout in Slide 21?